

Kaushik Vada

858-305-8647 | kaushikvada3@gmail.com | [linkedin.com/in/kaushikv198](https://www.linkedin.com/in/kaushikv198) | github.com/kaushikvada

Education

University of California, Riverside

Bachelor of Science in Electrical Engineering

GPA: 3.93/4.00 | University of California Regents Distinguished Scholar

Riverside, CA

Aug 2023 – June 2027

Relevant Coursework: Introduction to VLSI, Data Structures and Algorithms, Design and Architecture of Computer Systems, Machine Organization and Assembly Language, Digital Logic Design, GPU Architecture and Programming

Technical Skills

Programming Languages: Verilog, C/C++, Python, TCL

Tools: Synopsys (PrimeTime, PrimeTime PX, Design Compiler, Fusion Compiler, VCS), Cadence Innovus, Xilinx Vivado

Concepts: Physical Design, Floorplanning, Placement & Routing, Clock Tree Synthesis (CTS), Static Timing Analysis (STA), SDC Timing Constraints, Power Analysis & Optimization, RTL Design, Computer Architecture

Protocols: UART, APB, Valid/Ready Handshake

Experience

Physical Design Intern

Tenstorrent

September 2026 – Present

Santa Clara, CA

- Perform **static timing analysis (STA)** on die-to-die interconnects in a next-generation AI accelerator SoC with RISC-V cores, using **Synopsys PrimeTime** with post-route netlists, SPEF parasitics, UPF power intent, and SDC constraints.
- Analyze **multi-voltage timing paths** through level shifters connecting a dynamic-voltage SoC domain to a fixed-voltage PHY domain, evaluating voltage-dependent timing behavior.
- Extract corner-specific parasitics using **Synopsys StarRC** on Fusion Compiler databases and configure automated **RTL-to-GDS flows** from synthesis through route optimization.
- Investigate **AI/ML-based corner selection** targeting a reduction from approximately 46 PVT corners to 10, evaluating derating and guard-banding strategies to preserve timing coverage and reduce signoff turnaround.

SoC Physical Design Intern

Arm

June 2026 – September 2026

San Diego, CA

- Delivered a **tape-out candidate** for an active ARM AGI-CPU SoC block, incorporating **activity-driven power optimization** and two-pass placement optimizations and executing the **Cadence Innovus** flow from floorplanning through placement, CTS, routing, and post-route optimization; the resulting implementation was advanced for downstream STA and timing evaluation
- Compared multiple **FSDB activity profiles** to guide selection based on power and timing, with **activity-driven power optimization** yielding **18.4–19.5% lower reported power** versus baselines under matched workloads. Co-developed a two-pass placement flow with Cadence engineers, reducing estimated power by **5.86%** and cell area by **2.45%**.
- Reduced **measured mean RX data-path delay by 38.5%** using **Codex-assisted statistical modeling** of prior runs to guide setup-margin tuning, deliberately tightening constraints on positive-slack I/O paths to trigger optimization toward full-chip timing requirements.
- Enabled reusable **STA and PrimeTime PX workflows**, developing analysis scripts to compare implementation results, audit generated Tcl/YAML against applied constraints, and support international engineering teams.

Undergraduate Researcher — VLSI Systems & Computer Architecture Lab (VSCLab)

University of California, Riverside

September 2025 – April 2026

Riverside, CA

- Designing and implementing a custom open-source **RISC-V CPU core**, developing pipeline stages and verification infrastructure targeting full functional correctness.
- Applying **Verilog** and Synopsys **RTL-to-gate flows** in an active research environment, reinforcing skills in synthesis and gate-level simulation.
- Authoring and refining SDC timing constraints in **Synopsys Design Compiler** as part of the lab's constraint-driven synthesis workflow.

Projects

MiniGPU-2: Custom Graphics Accelerator ASIC

Mar 2026 – Present

- Implementing **floorplanning** and **place-and-route** in **Cadence Innovus** on a SAED 32nm checkpoint, iterating on timing and congestion for a custom ASIC.
- Worked with a team on **RTL-to-gate synthesis** in Synopsys Design Compiler and Cadence Genus for a SystemVerilog graphics pipeline spanning vertex transformation and rasterization.
- Designed dual-clock asynchronous FIFOs with Gray-coded pointer synchronizers to support communication across clock domains.